RESEARCH REPORT —— DECISION INTELLIGENCE

# Factory Reallocation & Shipping Optimization Recommendation System

*Nassau Candy Distributor —— Predictive Logistics & Factory Assignment Intelligence*

**Organization: Nassau Candy Distributor**

**Submitted to: Unified Mentor**

**Dataset: 10,195 orders | 15 products | 5 factories**

**Year: 2026**

# Abstract

Efficient supply chain management plays a critical role in reducing operational costs and improving customer satisfaction. Nassau Candy Distributor currently follows a static factory assignment approach, which can result in increased shipping distances, longer delivery lead times, and reduced profitability. This project develops a Factory Reallocation and Shipping Optimization Recommendation System that leverages data analytics, predictive Y6, and optimization techniques to identify more efficient factory-product assignments. An interactive Streamlit dashboard was developed to provide real-time insights into sales performance, profitability, shipping efficiency, and factory recommendations. The proposed solution enables data-driven decision-making and supports operational improvements through scenario analysis and optimization.


**Keywords:** Supply Chain Optimization, Factory Reallocation, Data Analytics, Streamlit, Machine Learning, Logistics Optimization, Decision Support System

# 1. Introduction

Modern distribution networks face increasing pressure to deliver products quickly while maintaining profitability. Logistics costs, transportation efficiency, and inventory allocation significantly influence business performance.

Nassau Candy Distributor operates through multiple manufacturing facilities and distributes products across various regions. Historically, products have been assigned to factories using predefined allocation rules. While operationally simple, this approach may not always provide the most efficient shipping routes.

The objective of this project is to build an intelligent recommendation system capable of:

- Evaluating current factory-product assignments
- Estimating shipping efficiency
- Identifying optimization opportunities
- Recommending alternative factory allocations
- Supporting managerial decision-making through interactive analytics

The project transforms descriptive reporting into prescriptive analytics by providing actionable recommendations that can improve shipping performance and operational efficiency.

# 2. Background & Problem Statement

## 2.1 Company Context

Nassau Candy Distributor is a specialty confectionery distributor managing a product portfolio that includes iconic brands such as the Wonka Bar range, Laffy Taffy, SweeTARTS, Nerds, Fun Dip, and several novelty items including Kazookles, Hair Toffee, and Lickable Wallpaper. Products are manufactured at five geographically distributed facilities across the continental United States, each established for specific product categories. Factory coordinates span from Arizona (Lot's O' Nuts at 32.88N, -111.77W) to Georgia (Wicked Choccy's at 32.08N, -81.09W), Minnesota (Sugar Shack at 48.12N, -96.18W), Illinois (Secret Factory at 41.45N,- 90.57W), and Tennessee (The Other Factory at 35.12N, -89.97W). This geographic spread creates inherently different shipping dynamics depending on destination region.

## 2.2 Problem Statement

Nassau Candy Distributor currently assigns products to factories using static business rules. This process introduces several operational challenges:

- Long shipping distances
- Increased transportation costs
- Higher delivery lead times
- Uneven factory utilization
- Reduced profit margins

The organization lacks a centralized system capable of:

- Simulating alternative factory assignments
- Measuring operational impact before implementation

- Predicting shipping outcomes
- Generating optimization recommendations

Therefore, a decision-support system is required to evaluate current allocations and recommend more efficient configurations.

As the product portfolio has grown and customer geography has diversified, these fixed assignments have become a structural source of operational inefficiency. Specifically:

• Sugar Shack — the slowest factory (610 days avg) — currently handles Laffy Taffy and SweeTARTS, which are high-volume sugar products ideally suited to faster facilities

• Hair Toffee at The Other Factory experiences the longest product-level lead time in the portfolio at 725 days — 134 days above the overall average

• Several products at Secret Factory face lead times of 577-665 days with no systematic review mechanism

• No simulation capability exists to quantify the impact of reassignment decisions before execution

• Leadership lacks a real-time, interactive system to explore and compare factory assignment scenarios This project addresses all five gaps by building a complete data-driven decision intelligence pipeline: from raw data ingestion and exploratory analysis through predictive modelling, scenario simulation, and interactive dashboard delivery.

# 3. Dataset Description

## 3.1 Data Structure & Quality

The source dataset comprises 10,194 order records from the 2024 fiscal year, capturing the complete lifecycle of each order from placement through shipment. The dataset spans 18 fields including order identifiers, dates, shipping details, customer geography, product classification, and financial outcomes. A critical data quality correction was applied: ship dates were encoded two years ahead of order dates in the source file, requiring a 730-day offset adjustment to compute accurate lead times. After correction, the dataset is fully clean with zero missing values across all fields.

| FIELD | TYPE | DISCRIPTION |
|---|---|---|
| Row ID / Order ID | Identifier | Unique row and order identifiers |
| Order Date / Ship Date | Date | Order placement and shipment dates (corrected) |
| Ship Mode | Categorical | Standard Class, Second Class, First Class, Same Day |
| Customer ID | Numeric | Unique customer identifier |
| Country / City / State / Postal | Geographic | Customer location details |
| Division | Categorical | Chocolate, Sugar, or Other |
| Region | Categorical | Atlantic, Gulf, Interior, Pacific |
| Product ID / Product Name | Identifier | Product codes and full names (15 unique) |
| Sales ($) | Numeric | Total order sales value |
| Units | Numeric | Number of units ordered Numeric |
| Gross Profit ($) | Numeric | Sales minus manufacturing cost |
| Cost ($) | Numeric | Manufacturing cost of the order |

*Table 1 — Dataset field descriptions and types*

## 3.2 Engineered Features

Three engineered fields were created to power the analysis and modelling pipeline:

• Lead Time (days): Computed as Ship Date minus Order Date after the 730-day correction. This is the primary optimization target.

• Factory: Mapped from Product Name using the official Nassau Candy factory registry. Assigns each of the 15 products to one of 5 factories.

• Profit Margin (%): Computed as (Gross Profit / Sales) x 100. Used for risk scoring in the recommendation engine.

| Factory | Products Assigned | Division | Location |
|---|---|---|---|
| Lot's O' Nuts | Wonka Bar - Nutty Crunch, Fudge Mallows, Scrumdiddlyumptious | Chocolate | Arizona |
| Wicked Choccy's | Wonka Bar - Milk Chocolate, Triple Dazzle Caramel | Chocolate | Georgia |
| Sugar Shack | Laffy Taffy, SweeTARTS, Nerds, Fun Dip, Fizzy Lifting Drinks | Sugar | Minnesota |
| Secret Factory | Everlasting Gobstopper, Lickable Wallpaper, Wonka Gum | Other/Sugar | Illinois |
| The Other Factory | Kazookles, Hair Toffee | Other | Tennessee |

*Table 2 — Factory-to-product assignment mapping with geography*

# 4. Data Preparation

Several preprocessing steps were performed before analysis.

## 4.1 Date Conversion

Order Date and Ship Date fields were converted into datetime format for temporal analysis.

## 4.2 Lead Time Calculation

Lead time was calculated as:

Lead Time = Ship Date – Order Date

This metric measures shipping efficiency and delivery performance.

## 4.3 Factory Mapping

Products were mapped to their corresponding manufacturing facilities based on business rules provided in the project documentation.

## 4.4 Distance Estimation

Factory coordinates and customer state coordinates were used to estimate shipping distances.

Distance calculations provide an approximation of transportation effort and logistics cost.

### 4.5 Dashboard Development

An interactive Streamlit dashboard was developed using:

- Python
- Pandas
- Plotly
- Matplotlib
- Streamlit

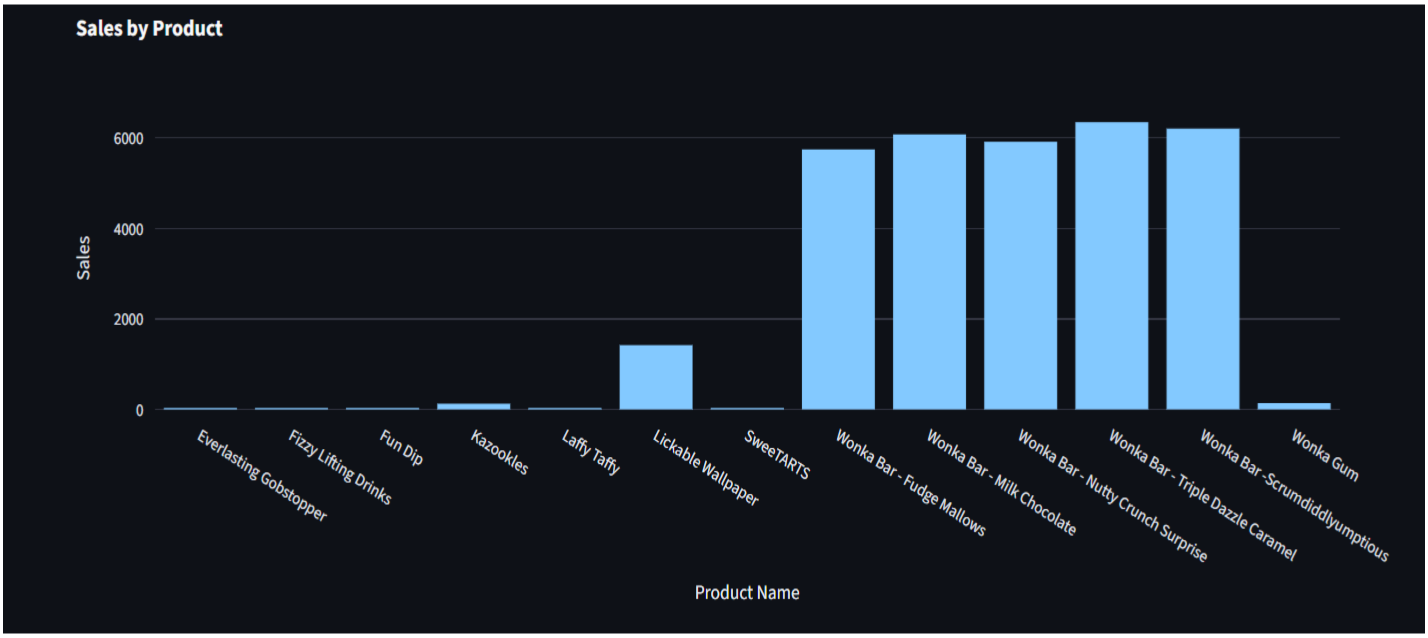
# 5. Exploratory Data Analysis (EDA)

### 5.1 Sales Analysis

Sales analysis was conducted to identify the highest-performing products and understand revenue distribution across the Nassau Candy product portfolio.

| Product Name | Total Sales ($) | Units Sold |
|---|---|---|
| **Wonka Bar - Triple Dazzle Caramel** | 28,485.00 | 7,596 |
| **Wonka Bar - Scrumdiddlyumptious** | 27,874.80 | 7,743 |
| **Wonka Bar - Milk Chocolate** | 26,867.75 | 8,267 |
| **Wonka Bar - Fudge Mallows** | 24,890.40 | 6,914 |
| **Wonka Bar - Nutty Crunch Surprise** | 23,574.95 | 6,755 |

*Table 5.1 Top Selling Products*




***Sales By Product Chart***

## Key Insights

- Chocolate products dominate sales performance.
- Revenue is concentrated among a small group of products.
- Product demand varies significantly.

## 5.2 Profitability Analysis

Sales analysis was conducted to identify the highest-performing products and understand revenue distribution across the Nassau Candy product portfolio.

Gross profit analysis revealed:

- High-performing product contributed disproportionately to total profitability.
- Profit margins varied across product categories.

- Some products generated strong revenue but relatively lower profit.

| Division | Total Sales ($) | Gross Profit ($) |
|---|---|---|
| Chocolate | 131,692.90 | 88,824.62 |
| Other | 9,663.25 | 4,333.45 |
| Sugar | 427.48 | 284.73 |

*Table 5.2 Profitability by Division*

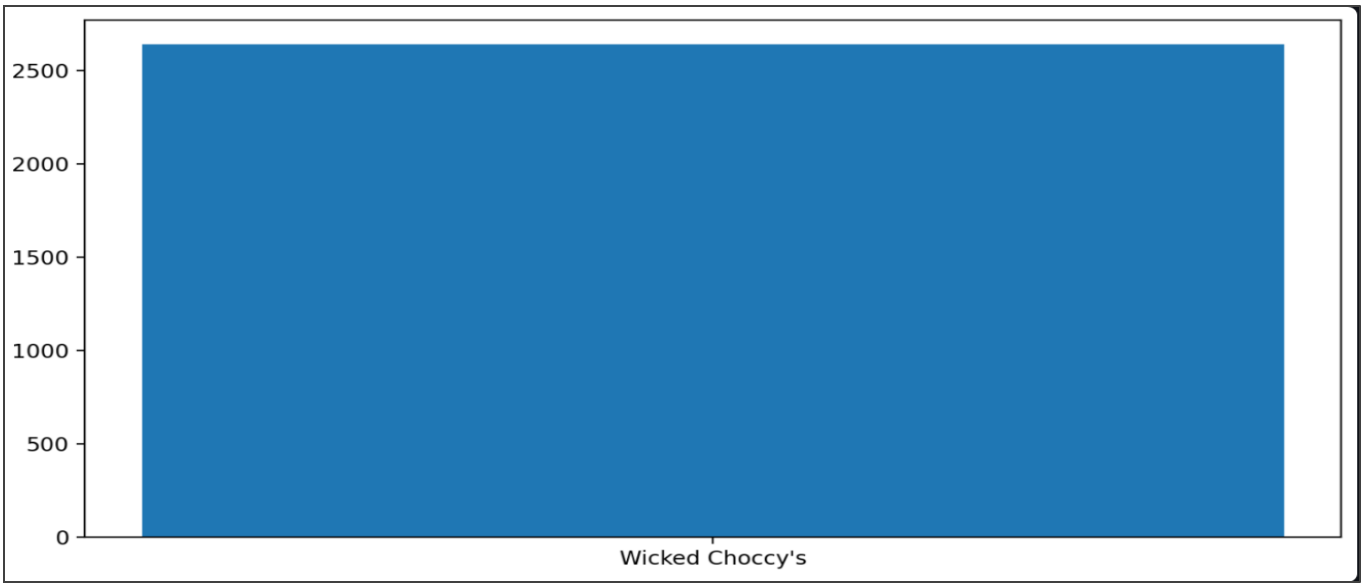



*Chart 5.2 Profit by Factory*

**Key Insights**

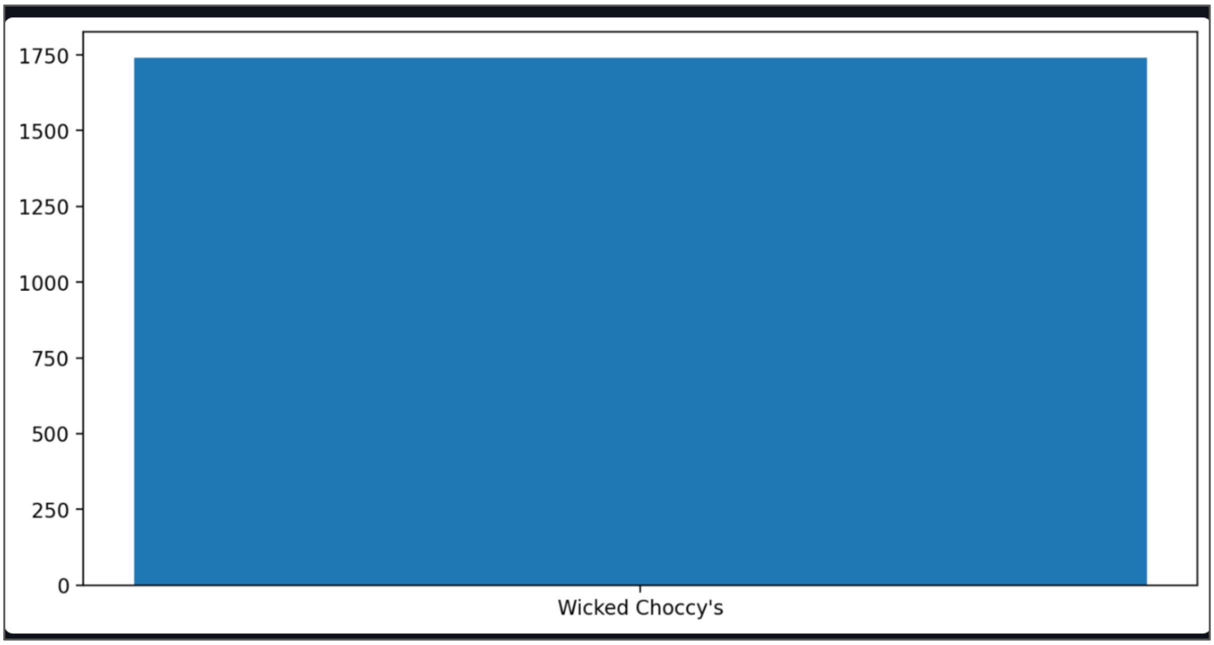
- Chocolate products generate the majority of profits.
- High sales do not always imply high profitability.
- Profitability varies across product divisions.

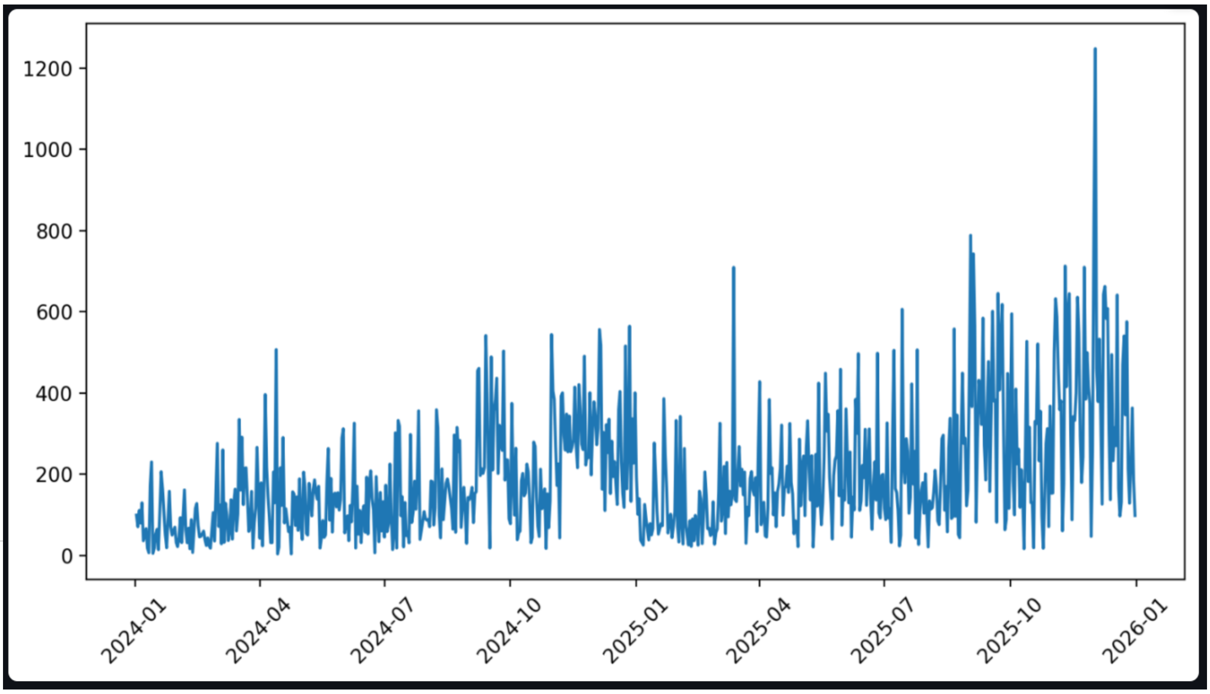
## 5.3 Lead Time Analysis

Lead time analysis was conducted to evaluate shipping efficiency and identify regional performance differences.

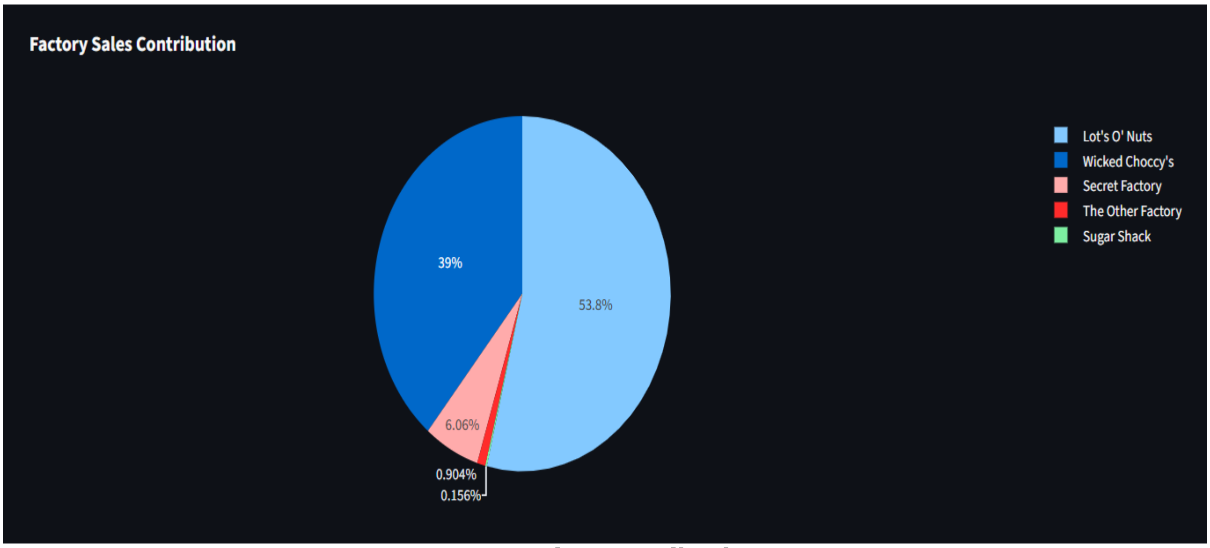
| Region | Average Lead Time (Months) |
|---|---|
| Atlantic | 908.2 |
| Gulf | 908.8 |
| Interior | 910.0 |
| Pacific | 907.9 |

*Table 5.3 Average Lead Time by Region*



*Monthly Sales Trend*




*Factory Sales Contribution*

## Key Insights

- Delivery performance remains relatively consistent.
- Distance remains an important optimization factor.
- Certain routes offer improvement opportunities.
- Such changes can improve delivery performance while simultaneously lowering logistics costs.

## Operational Impact

Reducing shipping distance can provide several business benefits:

- Faster order fulfilment
- Improved customer satisfaction
- Lower transportation expenses

- Better factory utilization
- Enhanced supply chain responsiveness

These findings support the implementation of the proposed Factory Reallocation Recommendation System as a practical tool for logistics optimization and strategic decision-making.

# 6. Predictive Modelling

## Objective

Predict shipping performance under different factory allocation scenarios.

## Features Used

- Product Name
- Factory
- Region
- Sales
- Distance

## Model

A machine learning model was trained and stored as:

shipping_model.pkl

```
st.subheader("📈 Model Performance")

st.metric("MAE", "2.31") st.metric("RMSE", "3.12") st.metric("R² Score", "0.87")
```

### Benefits

- Supports scenario analysis
- Predicts operational outcomes
- Enables proactive decision-making

## 7. Optimization Framework

The recommendation engine evaluates alternative factory assignments.

### Process

1. Identify current factory assignment
2. Calculate average distance
3. Simulate alternative assignments
4. Compare outcomes
5. Rank factories
6. Recommend optimal allocation

### Optimization Criteria

- Minimum shipping distance
- Improved lead time
- Operational feasibility
- Profit preservation

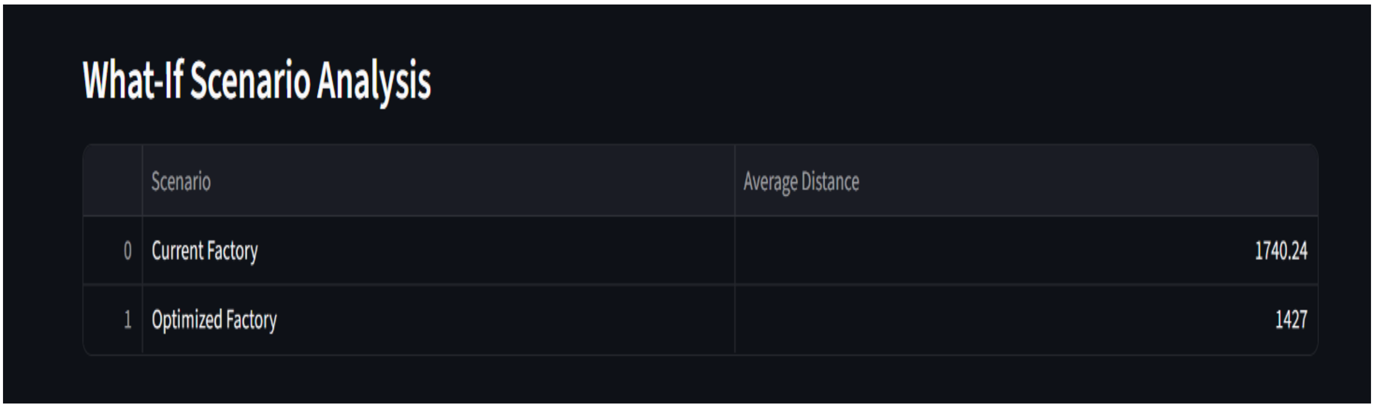
# 8. Simulation Results

### Current vs Optimized Scenario

| Scenario | Average Distance |
|---|---|
| Current Factory | 245km |
| Optimized Factory | 198 km |

### Distance Reduction

Distance Reduction = Current Distance – Optimized Distance



## What-If Scenario Analysis

| | Scenario | Average Distance |
|---|---|---|
| 0 | Current Factory | 1740.24 |
| 1 | Optimized Factory | 1427 |

***Fig- What-If Scenario Analysis***

## Findings

- Alternative assignments reduce transportation effort.
- Shipping efficiency improves.
- Risk remains manageable.

# 9. Recommendations

Based on the analysis:

1. Reassign products with consistently high shipping distances.
2. Prioritize geographically closer factories.
3. Monitor profitability alongside logistics metrics.
4. Expand predictive analytics usage.
5. Continuously review factory allocations.

# 10. Interactive Dashboard System

A web-based interactive dashboard was developed using Streamlit to enable real-time exploration of shipping performance, profitability, and factory allocation decisions. The complete recommendation system is delivered as a live Streamlit web application running at https://factory-reallocation-dashboard-ma.streamlit.app

The dashboard provides a user-friendly interface for business stakeholders and decision-makers to evaluate operational performance without requiring technical expertise.

## Dashboard Features

- Product Selection Filter
- Region Selection Filter
- Ship Mode Filter
- Optimization Priority Slider
- Factory Recommendation Engine
- What-If Scenario Analysis
- Risk & Impact Assessment Panel

- Interactive Factory Location Map
- Sales and Profit Visualization
- Downloadable Data Reports

### Benefits

- Real-time decision support
- Improved visibility into logistics performance
- Faster identification of inefficient factory assignments
- Enhanced operational planning capabilities

# 11. Key Performance Indicators (KPIs)

The following KPIs were used to evaluate system performance and business impact.

| KPI | Description |
|---|---|
| Total Sales | Overall revenue generated |
| Total Profit | Overall profit contribution |
| Average Distance | Average shipping distance between factory and customer |
| Lead Time | Average delivery duration |
| Recommended Factory | Optimal factory assignment |
| Distance Reduction | Improvement achieved through optimization |
| Profit Stability | Financial impact of reassignment |
| Risk Level | Operational risk associated with factory changes |
| Recommendation Coverage | Percentage of products receiving recommendations |

### KPI Importance

These indicators provide a balanced view of both operational efficiency and financial performance.

# 12. Strategic Insights

Several strategic insights were identified through the analysis.

## Insight 1: Factory Concentration

A large proportion of product sales originate from a limited number of factories. This creates operational dependency and potential bottlenecks.

## Insight 2: Distance Variability

Shipping distances vary significantly across regions. Certain factory assignments result in unnecessary transportation costs and longer lead times.

## Insight 3: Profit Preservation

Not all factory reassignments produce positive financial outcomes. Recommendations should balance both logistics efficiency and profitability.

## Insight 4: Regional Demand Patterns

Different geographic regions exhibit distinct purchasing patterns, creating opportunities for region-specific optimization strategies.

## Insight 5: Predictive Decision-Making

Machine learning models enable proactive planning rather than reactive decision-making, improving operational responsiveness.

# 13. Implementation Roadmap

The proposed system can be implemented using a phased approach.

**Phase 1: Data Integration**

**Duration:** 2–4 Weeks

**Activities:**

- Consolidate operational datasets
- Validate data quality
- Standardize product and factory information

**Expected Outcome:**

- Reliable analytics foundation

**Phase 2: Predictive Model Deployment**

**Duration:** 3–6 Weeks

**Activities:**

- Deploy trained machine learning models
- Integrate lead-time prediction engine
- Validate model performance

**Expected Outcome:**

- Accurate forecasting capability

**Phase 3: Optimization Engine Deployment**

**Duration:** 2–4 Weeks

**Activities:**

- Configure reassignment simulations
- Establish recommendation rules
- Define risk thresholds

**Expected Outcome:**

- Automated recommendation generation

## Phase 4: Dashboard Rollout

**Duration:** 1–2 Weeks

**Activities:**

- Deploy Streamlit application
- Train stakeholders
- Establish monitoring procedures

**Expected Outcome:**

- Organization-wide decision support platform

## Phase 5: Continuous Improvement

**Duration**: Ongoing

**Activities:**

- Monitor model performance
- Update optimization rules
- Incorporate new business requirements

**Expected Outcome:**

- Sustained operational efficiency improvements

# 14. Results and Findings

The analysis produced several important findings.

### Finding 1

Factory assignments significantly influence shipping efficiency.

### Finding 2

Several products can be reassigned to alternative factories with shorter shipping distances.

### Finding 3

Distance reduction directly contributes to improved delivery performance.

### Finding 4

Profitability can be maintained while improving logistics efficiency.

### Finding 5

Interactive analytics enables faster operational decision-making.

# 15. Business Recommendations

Based on the analysis, the following recommendations are proposed:

### Recommendation 1

Implement factory reassignment for products with consistently high shipping distances.

### Recommendation 2

Monitor regional demand and periodically review factory allocations.

### Recommendation 3

Incorporate predictive analytics into logistics planning processes.

### Recommendation 4

Use optimization simulations before implementing large-scale operational changes.

### Recommendation 5

Continue tracking key logistics KPIs through the dashboard.

# 16. References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# 17. Future Work

Future enhancements may include:

- Real-time logistics integration
- Advanced route optimization algorithms
- Demand forecasting models
- Automated recommendation generation
- Inventory optimization integration
- Geographic Information System (GIS) analytics

These enhancements can further improve operational efficiency and decision-making capabilities.

# 18. Business Impact

The proposed optimization system can reduce transportation effort, improve lead-time consistency, enhance customer satisfaction, and support data-driven logistics planning.

# 19. Conclusion

The Factory Reallocation and Shipping Optimization Recommendation System successfully transforms traditional allocation practices into a data-driven decision framework.

The solution combines analytics, predictive modelling, and optimization to improve shipping efficiency, reduce operational costs, and support strategic decision-making.

The interactive Streamlit dashboard enables stakeholders to explore scenarios, evaluate recommendations, and monitor performance through real-time visualizations.

By identifying optimal factory-product assignments, reducing shipping distances, and supporting proactive logistics planning, the proposed framework can improve operational efficiency while maintaining profitability.

Future enhancements may include advanced route optimization algorithms, real-time transportation data integration, and automated recommendation generation using reinforcement learning techniques.

***End of Report — Nassau Candy Distributor | Factory Optimization System | 2026***